TECHIN 510 | Veronika S.P. | 4/6/26

Submission

Component A Deliverables

1. Interview notes

- Direct quotes -- write down 1-2 exact phrases the Coordinator uses
    a. “So far my issue is whether I can make a better solution then just editing an excel sheet. I have deadlines each week for students to submit an order.”
    b. “The excel sheet enables me to see who was submitted, what they want and how many. In the end I would like something easier for students to submit and then spit out some form of .xlsx so that all that information can be put into a cumulative excel sheet that tracks each spending.”

- Surprises -- what did you not expect to hear?
    c. I would not like to use an app as I would like to utilize a computer with a larger screen to keep track of all the data I am inputting.

- Current workarounds -- how is the problem being handled today (spreadsheets, email, memory, signs)?
    d. Spreadsheets

- What success looks like -- in the interviewee's own words

    e. “In the end I would like something easier for students to submit and then spit out some form of .xlsx so that all that information can be put into a cumulative excel sheet”
    f. All-in-one area workspace, so she can see a lot at once

Observe the User Journey:

Timeline	How long does this step take?	Where does the information live?	What could go wrong at this step?
1. Student submits an order/purchase request	minutes	Excel	Students unsure how to write information down format wise. Missing information.
2. Instructor reviews for approval	minutes	Excel	Instructor approves but doesn’t communicate any missing information.

			Instructor refuses order request.
3. Dorothy reviews the submission for any missing information	minutes	Excel	Not sufficient info online to fill in the missing information on the spreadsheet. Can't purchase any needed items in time.
4. If any info is missing Dorothy will follow up with the student via online messaging	Minutes to days	Teams messaging or email	Delayed communication, missed opportunity to get the order delivered in time
5. Student and Dorothy communicate and resolve the situation	Minutes to days	Teams messaging or email	Delay in communication. Students can accidentally send the wrong info again.
6. Dorothy logs her information on the spreadsheet and keeps track of the orders	Minutes	Excel	Delivery time can be longer than expected. Students have to ask for delivery status; there's no automatic notification.

2. Problem statement: 1 sentence in the template

When students submit to check if their supplies were approved, they currently check the online spreadsheet on their own occasion or wait for a response from instructor or coordinator, which causes delay or mishaps during a tight project timeline whenever coordinator is messaging students by group.

Component B Deliverables

Reflection Deliverables

Before you leave, take a few minutes to write 3-5 sentences addressing the following questions. Submit these as part of your Component B deliverables.

1. What surprised you about AI-assisted coding? Was it easier or harder than you expected? What moment made you think "wow" or "wait, that is not right"?
    - I went wow, I'm cooked because I will now feel like a fraud and not a genuine computer programmer. I think people who majored in computer science before AI have the engineering skills to optimize code even with what AI provided because they have the background knowledge to actively understand code. While for me, it's like "Chegging" answers, it does not feel like active learning.

2. What did the AI get wrong? Did you encounter errors, unexpected behavior, or code that did not match what you asked for? How did you fix it?

The standard format for accessibility did not pass. Therefore graphics need to get tailored to meet the 4.5:1 ratio. The user-interface set-up like buttons or navigation pane does not look like the typical one from a company website, that needs to be fixed. So it's all graphic design mistakes that can be easily tailored to fit industry standards.

3. Could you explain your code? Pick one section of your app -- could you walk a classmate through what it does, line by line, without looking at Cursor's explanation?

The easiest thing I understand and probably can only explain is setting up the "database" with a class, storing all the desired inputs to request a purchase for items.

```
class LineItem:
    """One line on an order: what to buy, from whom, links, and
instructor/coordinator workflow flags."""
    item_id: str
    name: str
    quantity: float
    unit_price: float
    total_price: float
    supplier: str
    project_link: str
    instructor_approval: str = "pending"  # pending | approved |
rejected
    ordered: bool = False
    estimated_delivery: str = ""
```

4. What did you learn from the interview? How did talking to Dorothy (Program Coordinator, Student Purchasing) change what you built, compared to what you might have built without that conversation?

I learned that user research is important. That will help define the main objective and know when a project is “good enough”. I didn’t know she wanted to keep the Excel format. Some people want new and better systems, others want to optimize what they already have.

5. AI Usage Log: Document at least 3 interactions with AI tools (include Cursor prompts from build/iteration). For each, write:
    - the prompt you gave
    - what the AI produced
    - whether it worked on the first try, and what you fixed if it did not

1. INTERACTION 1, Asked for:

Build a Streamlit app called “GIX Student Purchasing” for Students who need to order supplies for their school project. There is a form they need to fill out, student name, “+ add item” button for every item they want in their list and when you press it student fills in the item name, quantity, unit price, total price, supplier and link where the project is located. And a table is filled out with the same information where the instructor and check it off for approval (check yes for approved, no to reject it). The coordinator uses the table to "checkmark" that it’s been ordered with estimated delivery date, and that notifies the student via email that order is approved. If the student needs to make returns or cancel the order there should be a button option to make edits. For every submission whether it’s the student, instructor or coordinator each person gets an email to notify them of what the action was. Make sure the layout is simple to follow, easy to use.

2. What Cursor produced:

One python file with a .gitignore file. Created a Streamlit app with JSON persistence, role-based tabs, and SMTP email notifications. Has variables for users submission and ID and input fields like name, quantity, item name, unit price. A view for student, instructor and coordinator. Instructor to approve pend, and coordinator to review and provide estimated delivery date. Cursor also produced a view of the activity log.

3. Whether it worked on the first try, or what you had to fix:

It worked the first try, I did want to tailor the user interface. Students shouldn’t view the staff page only the student page. Sign in as a student, instructor or coordinator. The SMTP isn’t configured because there isn’t an official server set up to send emails. And the total price should be automatically updated based on quantity and unit price specifications.

1. INTERACTION 2, Asked for:

I notice you went from Title straight to subheader and plain text. Is there a way where you can have titles, headers and then subheaders?

2. What Cursor Produced:

There is title → Student or Staff Portal (header) → Subheaders for purchase request, tracking changes and sections inside the form

3. Whether it worked on first try, or what you had to fix

1. INTERACTION 3, Asked for:

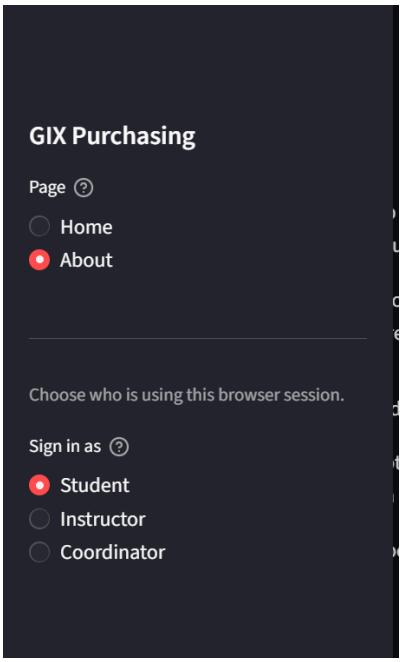
"Add a sidebar with a radio button that lets the user switch between 'Home' and 'About' pages. The About page should show a description of the project and a link to the GIX website."

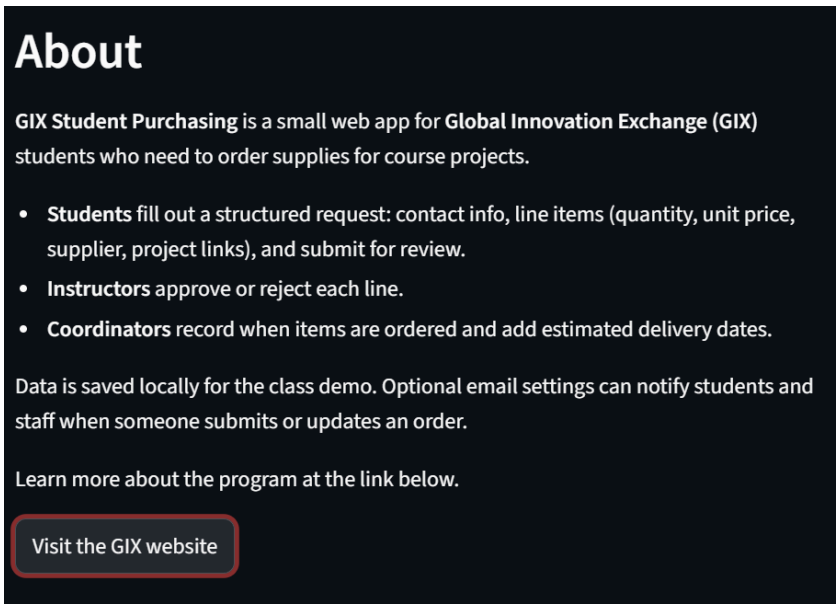
2. What Cursor Produced:

A page radio at top with Home and About and a divider “Home | About”. The About page has a link with the GIX website, it opens in a new tab when clicked.

3. Whether it worked on first try, or what you had to fix

It works, but the formatting and wording can be better. The button formatting looks different than what I typically see for a website:





1. INTERACTION 4, What I asked for:

build a streamlit app based on this prompt:

The Problem

New GIX students can't find campus resources — the makerspace, bike storage, free printing, quiet study nooks. No single digital resource exists; students rely on word of mouth and outdated PDFs. Build a wayfinding prototype that solves this.

This component is required and part of your Week 1 submission.

What You Build

A Streamlit app with a set of campus resources stored in a data structure you design

Search and filtering functionality

A display showing matching resources with relevant details

I took a guess on how the code would work based on my I/P/O chart. Let me know if I got idea down

input: ISearch (text input) Filter (select multiple choice) Button to enter search

process: Match the words to the database based on filter and text search for any similarities. LLM to assist in nuance words for similarities

Output: Display resources with the closest match amnd relevant details Or Dislay “sorry we couldn’t find what you were looking for :(“

I can provide vague info on where everything is:

laser cutter, 3D printers, PCB lab all in the maker space

classes in the main conference room

program meetings on the 1st floor

phone rooms - 1st and 2nd floor

food options - cafe, bar just around the street

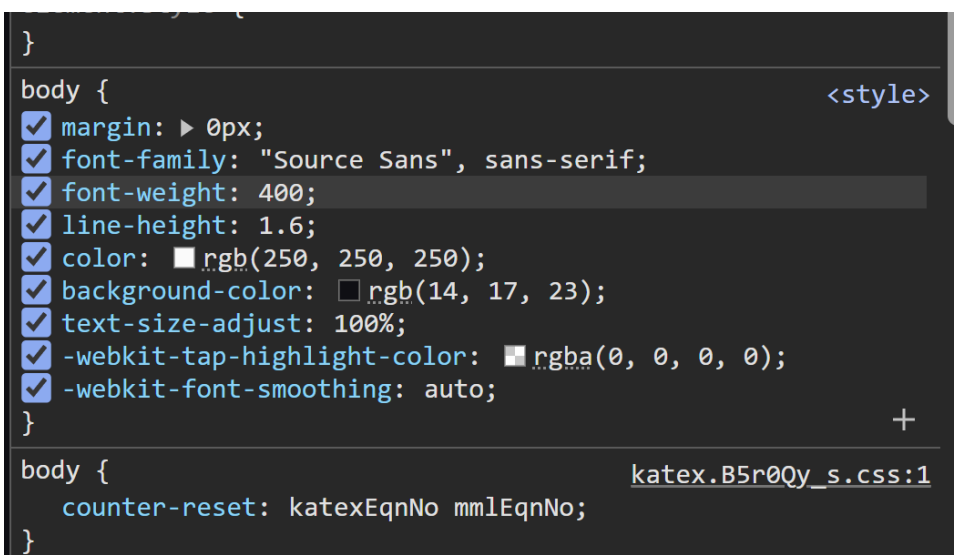
here is the institute with some background info: https://gix.uw.edu/

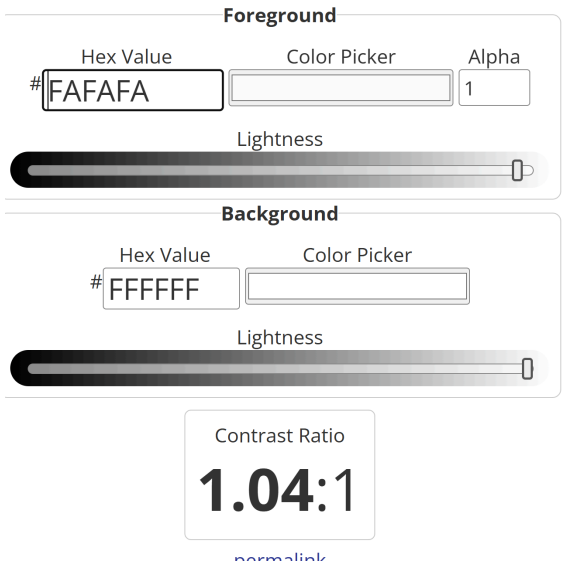
2. What Cursor produced:

Match text against a small Python “database” a list of dictionaries. Use lightweight fuzzy matching as stand in for LLM nuance since LLM isn’t required.

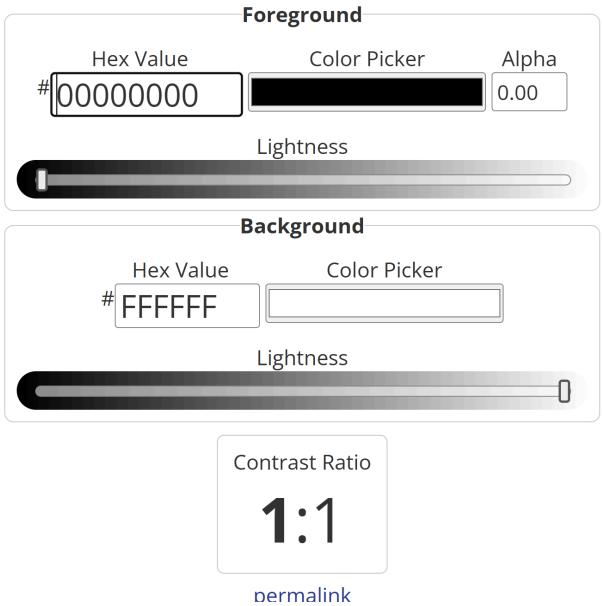
3. Whether it worked on the first try: yes, no fixes

6. Accessibility baseline results: Briefly report:
   - color contrast check result (pass/fail and any fix)
     i. I’m going to be real. I didn’t have the energy to fix the font colors to fit the standard. Just know it failed.
   - semantic headings check result (pass/fail and any fix)
     i. Headings jumped from Title to subheading. Had Cursor fix the hierarchy to include heating before subheading.





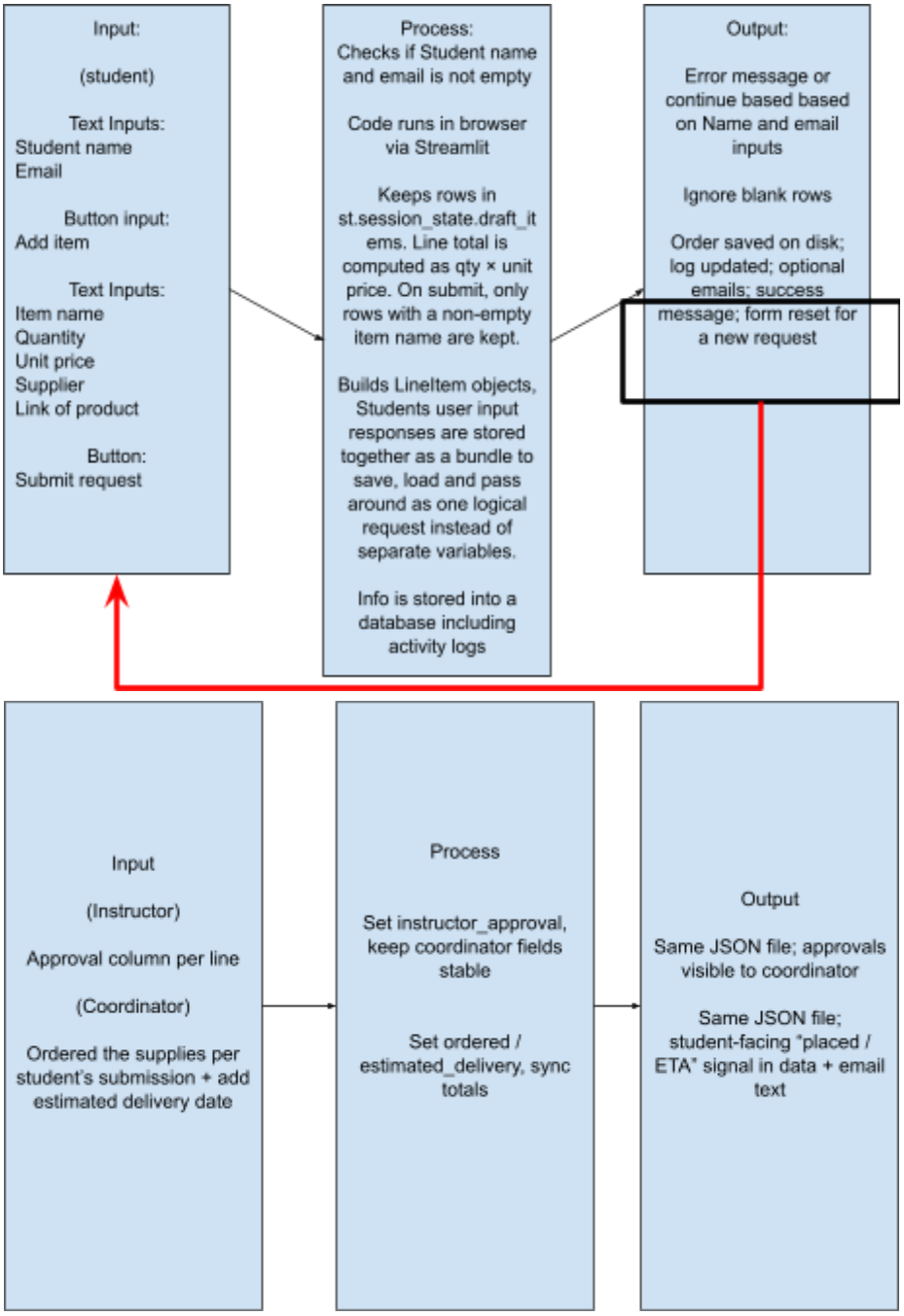
Fails



Fails

Component C Deliverables

1. Architecture diagram



2. Design Decision Log

Field	Your Entry
Decision	What did you decide? I decided to make a website. Make

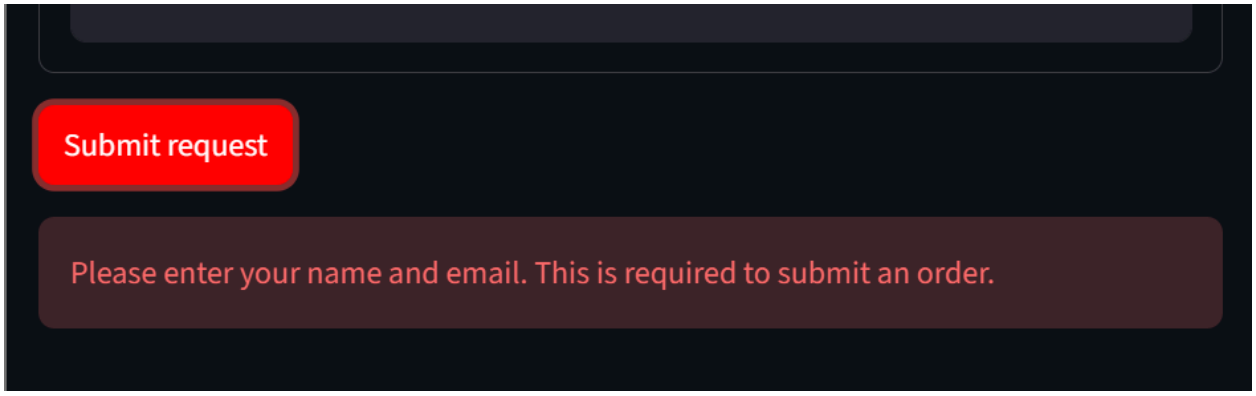
	an online form to submit purchase requests
Alternatives considered	What else could you have done? If I was there for the interview, I would've kept the Excel format because that's what the interviewee was requesting. And optimized it with features that would make it helpful for all users (students/staff)
Why you chose this	What constraint drove it? I chose this decision so that there were automated tasks to make filling out and submitting requests seamless. One of the ideas was to notify users of submission logs and estimated delivery dates.
Trade-off	What did you give up? An online form can look clunky, and you can view all information on one page. Have to go through a separate page and some clicks.
When would you choose differently?	Under what conditions? Under the condition my client doesn't like the design I would choose differently. Under the condition, the user interface prints errors and the system is no longer seamless.

Component D Deliverables

1. Validation results (the completed smoke-test table)

#	Feature Tested	Action You Took	Expected Result	Actual Result	Pass/Fail
1	*Text input for student name and email*	*Veronika* *vpon@uw.edu*	*Should show up in the submission*	Code is fixed. Name shows up on the grid after submission as the first cell of the row.	*Pass*
2	*Submit request button*	*Submit without adding any items*	*Message should show "Add at least one item with a name"*	*Message showed "Add at least one item with a name"*	*Pass*
3	*Button to switch between Home and About page*	*Load the About page*	*Description on the purpose of the website should show up*	*Description on the purpose of the website showed up*	*Pass*

2. Screenshot of your running app showing at least one tested feature



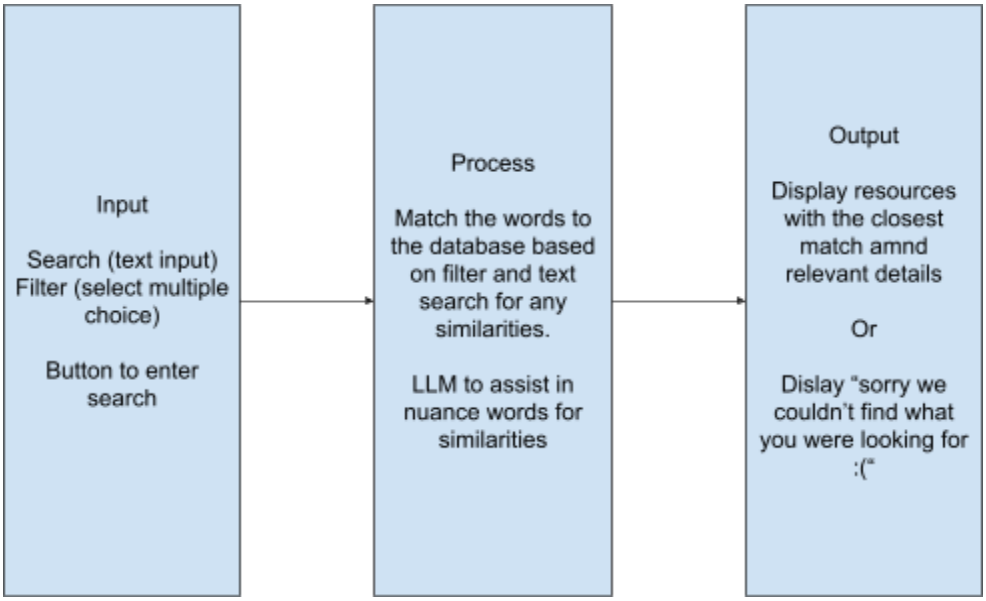
3. Quality gate checklist (completed)

- ☐ Smoke test table completed with 3 tested features
- ☐ Any failed test is either fixed and re-tested, or clearly documented
- ☐ Screenshot of the running app included

- [ ] Accessibility baseline results recorded (color contrast + semantic headings)

Component E Deliverables

1. Wayfinder app implementation in your repo (code and data structure)
2. I/P/O diagram for the Wayfinder app



3. Edge-case validation notes for 2 edge cases (with why they matter)

Edge case 1: Empty search text with only filters selected

What I tested: Left the text input blank and selected only Makerspace in the category filter, then clicked Search.

Why it matters: New students might not know exact keywords (e.g., “laser cutter”), but they may know the general type of place (e.g., “makerspace”). The app should still show relevant results instead of nothing or an error when the text box is empty.

Edge case 2: Search text that doesn’t match anything + filter

What I tested: Searched for "swimming pool" while selecting all categories and then clicked Search.

Why it matters: Users will sometimes search for resources that don’t exist at GIX. The app should respond clearly (e.g., with “Sorry, we couldn’t find what you were looking for :(") so they understand it’s a valid search but there is no matching resource.

4. One `assert` statement for data integrity (include it in code and mention where)

```
if __name__ == "__main__":

    # Simple data integrity check

    resources = build_resource_data()

    names = [r["name"] for r in resources]

    assert len(names) == len(set(names)), "Duplicate resource names
found in build_resource_data()"



    main()
```

5. Prompt log excerpt: initial prompt + one refinement, with a brief note on what changed and why

<u>Initial AI prompt (copy):</u>

"Build a streamlit app based on this prompt: The Problem… New GIX students can't find campus resources… Build a wayfinding prototype… A Streamlit app with a set of campus resources stored in a data structure you design, Search and filtering functionality, A display showing matching resources with relevant details… I took a guess on how the code would work based on my I/P/O chart… I can provide vague info on where everything is… [resource notes] … here is the institute with some background info: https://gix.uw.edu/"

<u>Refinement prompt (copy):</u>

"It seems multiple choice overrides the typed in word search. I want them to work together to find all possibilities, not override one over the other."

<u>What changed and why:</u>

In the initial prompt, I asked the AI to build the full prototype app (data structure, search/filter inputs, and results display). After testing the app, I noticed a usability issue: the category filter (multiple choice) was acting like a strict gate and could hide relevant results even when the text search matched well. In the refinement, I focused the AI on improving the search logic so the text input and category filter combine to broaden results and rank them, instead of one input overriding the other. This makes the wayfinder more helpful for real users who may use both inputs at the same time.